TierPulse: Tier 1–3
Performance Management

Data Operating Theory → governed metric capture, structured escalation, and Tier 3 roll-up

J&J Innovative Medicine — LVV Raritan | TierPulse

Situation: Before TierPulse

Performance management was fragmented — no Tier 1, inconsistent Tier 2, and zero standardized Tier 3 leadership visibility

- **Tier 1 did not exist** — no daily operational pulse at the shop floor
- Tier 2 dashboards existed in silos — no unified escalation or action tracking
- Escalations lacked consistent owner, due date, and status capture
- No standardized Tier 3 leadership view across LVV sites
- Manual data pulls, no governed metric definitions, no auditability

Design intent: Real-time operational pulse + structured escalation + transferable roll-up. Not dashboards for dashboards' sake — decision infrastructure.

Conceptual Alignment: SQDCP Framework

Every metric at every tier maps to one of five performance pillars

SAFETY
- Operator safety
- Data integrity risk
- Compliance risk
- Error prevention

QUALITY
- Product quality
- Data accuracy
- Right-first-time performance

DELIVERY
- Cycle time
- Flow and throughput
- Schedule adherence

COST
- Resource utilization
- Rework and downtime
- Material waste

PEOPLE
- Training completion
- Engagement and adoption
- Retention and usability

Objectives: What We Set Out to Build

A governed, hybrid, transferable performance management system across all three tiers

PRINCIPLE	TARGET
Hybrid by design	Human confirms attainment; automation supplies evidence + trends
Governed	Metric catalog + definitions + auditability (who / when / what changed)
Transferable	Site configuration layer + standardized Tier 3 roll-up template
Escalation-driven	Structured action log: owner + due date + status at every tier

Execution: Analytical Ask by Data Operating Bucket

Six data operating buckets map directly to TierPulse outputs

DATA OPERATING BUCKET	TIERPULSE OUTPUT
1) Governance & Provenance	Maximize definition clarity: owner, target, frequency, status
2) Generation	Minimize manual pulls — source COMET / SAP / MES / Curve signals
3) Collection & Standardization	Maximize harmonization: canonical metric IDs, SQDCP mapping
4) Aggregation & Enrichment	Optimize enriched outputs: 6-day & 14-day trends, 7-day windowing
5) Analysis	Maximize decision support: persistent red detection, cross-functional flags
6) Application	Optimize tier-ready views: Tier 1 board, Tier 2 trends, Tier 3 roll-up

Execution: What We Built

Artifact-driven implementation — tangible outputs at every tier level

TIER 1 — DAILY BOARD

- Shop-floor operational pulse
- SQDCP metric status at a glance
- Owner-assigned escalation capture

TIER 2 — 2X/WEEK BOARD

- Trend-enriched metric views
- 6-day and 14-day trend signals
- Cross-functional flagging

UNIFIED ACTION LOG

- Owner + due date + status
- Escalation tracking across tiers
- Audit trail for governance

Scope: Current state is descriptive/diagnostic with structured escalation logic. Predictive and prescriptive capabilities are phased after automation maturity.

Execution: Data Integrity & Adoption

Guardrails and human-centered design ensure trust and sustained adoption

3ES FRAMEWORK

- Efficiency — streamlined workflows
- Effectiveness — better decision signals
- Experience — user-impact focus

DATA INTEGRITY CONTROLS

- Missing data guardrails
- Incorrect data detection
- Imprecise / irrelevant filters
- Counterserendipity checks for automation

Principle: TierPulse is designed for behavior-aligned adoption — balancing technical completeness with user experience. Standards prevent negative effects as automation scales.

Verified Results & Early Signals

Measurable impact from TierPulse implementation

83%

Reduction in report development time

80→90%+

On-time training completion rate improvement

- Faster overdue non-conformance identification through persistent red detection
- Improved material readiness visibility across production lines
- Consistent tracking and feedback loop sustaining performance gains
- Feedback-driven improvements showing maturity in operating model adoption

Roadmap: Tier 3 & Prescriptive Analytics

End-state target — automated, prescriptive Tier 3 roll-up for leadership decision queues

HISTORICAL — FOUNDATION

- Map Tier 1–3 process end-to-end
- Extract metric entities and linkages
- Automated concept chain extraction from historical boards

DIAGNOSTIC — QUALITY CONTROL

- Identify missing details and wrong information
- Rate comprehension and accuracy of historical metrics
- Link scores to outcomes and adverse events

PREDICTIVE — AUTOMATION

- Automate metric capture for 80%+ of Tier boards
- Signal generation mimicking high-quality historical patterns

PRESCRIPTIVE — TIER 3 ROLL-UP

- Portfolio-level leadership decision queue
- Process enhancement recommendations
- Cross-site comparability and benchmarking

Next Steps & Asks

Concrete actions to scale TierPulse from pilot to enterprise capability

- **Ask 1:** Evaluate further data enrichment via Databricks / COMET connection for automated signal generation
- **Ask 2:** Pilot Tier 3 roll-up board — monthly leadership cadence with decision queue
- **Ask 3:** Approve replication approach — site configuration layer to Sassenheim and beyond
- **Ask 4:** Establish data quality guardrails — minimum standard + ownership accountability

Parking lot: Metric catalog, meeting standards, source-system mapping, and FAQ documented in appendix.

Why This Matters

- TierPulse operationalizes data operating theory — not dashboards, but decision infrastructure
- Governed metric capture replaces manual pulls with automated, auditable signals
- Structured escalation drives accountability at every tier level
- Transferable design enables replication across LVV sites and beyond
- Tier 3 prescriptive roadmap positions leadership for portfolio-level decision-making

> **Data operating theory becomes real at the point where operational decisions are made.**